

Answers

Q	Ans	Q	Ans	Q	Ans
1	b	2	b	3	d
4	a	5	b	6	d
7	a	8	c	9	c
10	c	11	a	12	b
13	b	14	b	15	c
16	a	17	c	18	d
19	b	20	a	21	c
22	a	23	d	24	b
25	d	26	c	27	b
28	d	29	a	30	b
31	a	32	d	33	c
34	d	35	d	36	a
37	d	38	d	39	a
40	b	41	d	42	a
43	c	44	b	45	c
46	a	47	d	48	b
49	c	50	c	51	d
52	a	53	c	54	b
55	a	56	a	57	b
58	a	59	b	60	b

Selected working

(11) The East Gate Center (Harare) copies the mound architecture of termites, which have chimneys keeping temperature and humidity constant.

(16) Mosquitoes take blood through a needle-like proboscis; engineers copied this structure for the injection needle.

(25) Biotechnology “utilizes biological systems to develop useful products”; radioactive tracers, gears and circuits are not biological systems.

(40) Human Chorionic Gonadotropin (HCG) is produced in the placenta about six to ten days after fertilization; insulin, adrenaline and melatonin are different hormones.

(41) In the pregnancy urine test, one line is the control line (negative); two lines confirm pregnancy.

(45) Computer Information Technology Scanning (CITS) investigates information on diseases or cancerous areas of the human body.

(56) Trees “function in decentralized forms without a centralized order-providing unit (brain)”, so centralized, artificial and computerized forms are wrong.

(57) Caring for plants is an ethical treatment and planting trees is a national responsibility; all other options are listed in the textbook as unethical treatments.

(59) The three Rs are Reduction, Refinement and Replacement; the waste-management 3Rs (Recycle/Reuse) are not the principles given in the textbook.

(60) Replacement substitutes animal experiments with cell cultures, biosynthesis and computer simulated models; the other statements describe Reduction and Refinement.